

Yixiao Wang

+1-919-201-8680 | yixiao.wang@duke.edu | [yixiao-wang-stats.github.io](https://github.com/yixiao-wang-stats) | [Yixiao-Wang-Stats](https://github.com/Yixiao-Wang-Stats)

Durham, NC 27705, United States

EDUCATION

- **Duke University** Aug. 2024 – May 2026
M.S. in Statistical Science
Durham, NC, United States
 - GPA: 3.95/4.00
- **University of Science and Technology of China, School of the Gifted Young** Sep. 2020 – Jul. 2024
B.S. in Mathematics
Hefei, Anhui, China
 - Top 10% in School of the Gifted Young and the Department of Mathematics; **Outstanding Graduate**
 - Thesis: *Trimmed Mean for Partially Observed Functional Data* (Advisor: [Prof. Xiaohong Lan](#))

RESEARCH INTERESTS

- **Interpretable and theoretically grounded machine learning**, focusing on scalable optimization and structure learning methods that unify interpretability, performance, and practicality.
- **Theory-driven deep learning algorithms**, developing efficient, simple, and high-performance approaches for generative and multimodal systems, grounded in rigorous stability and approximation analyses.
- **Deep learning theory**, uncovering the underlying principles and mechanisms of modern architectures to explain their generalization, reasoning, and representation behaviors from theoretical and geometric perspectives.

RESEARCH EXPERIENCE

- **Duke University, Interpretable Machine Learning Lab** Jun. 2025 – Present
Supervisor: [Prof. Cynthia Rudin](#) Durham, NC, United States
 - **Developed near-optimal algorithms for sparse regression trees in seconds**, unifying constant-leaf and linear-leaf formulations under a dynamic-programming lookahead framework.
 - **Established computational complexity bounds and proved approximation guaranties**, showing that the proposed tree-optimization algorithms achieve solutions provably close to the global optimum.
 - **Built an optimized C++/Python package** with reproducible benchmarks on 10 UCI datasets, demonstrating consistent order-of-magnitude speedups over optimal baselines (OSRT, STreeD) while preserving accuracy.
- **Duke University, Department of Computer Science** Jun. 2025 – Present
Supervisor: [Prof. Anru Zhang](#) Durham, NC, United States
 - **Co-first-author, NeurIPS 2025 Workshop on What Can't Transformers Do? (Oral)**: Established the first rigorous in-context learning theory for time series forecasting, showing that Transformers are fundamentally suboptimal: proved a strictly positive excess-risk gap over linear predictors, quantified it with an inverse-linear convergence rate, and demonstrated exponential error compounding under Chain-of-Thought rollout.
 - **Co-first-author, ICLR 2026**: Modeled hidden-state trajectories of Transformers using Menger curvature to analyze logical consistency, uncovering geometric invariants that govern reasoning behavior.
- **Duke Center for Computational Evolutionary Intelligence (CEI)** Jan. 2025 – Present
Supervisor: [Prof. Yiran Chen](#) Durham, NC, United States
 - **Developed ZEUS**, a controllable training-free diffusion acceleration framework that achieves up to **3.2× end-to-end speedups** (2.0×–3.2× across SD-2, SDXL, and Flux) while improving perceptual fidelity ($\text{LPIPS} \leq 0.08$, $\text{FID} \leq 3.5$) over prior training-free methods.
 - **Co-first-author, ICML 2025**: Developed **SADA**, a stability-guided, training-free adaptive diffusion accelerator that achieves $\geq 1.8\times$ generation speedups across SD-2, SDXL, and Flux, while maintaining perceptual quality ($\text{LPIPS} \leq 0.10$, $\text{FID} \leq 4.5$).
 - **Contributing author, AAAI 2026**: Provided theoretical analysis for **FlashSVD**, deriving memory–compute complexity bounds and characterizing rank-dependent speedup and compression thresholds for low-rank Transformer inference.
- **Wake Forest University, Department of Computer Science** Jan. 2025 – Present
Supervisor: [Prof. Aditya Devarakonda](#) Winston-Salem, NC, United States
 - **Co-first-author NeurIPS 2025**: Proposed and led the development of **Enhanced Cyclic Coordinate Descent (ECCD)**, a Taylor-corrected block coordinate descent method for generalized linear models. The algorithm reduces repeated link-function evaluations and achieves provable improvements in computational efficiency and error bounds.
 - **Developed a high-performance C++-based R package** with optimized single-point and pathwise λ solvers, supporting large-scale regularized regression with strong numerical stability.
 - **Demonstrated 2–4× speedups** over GLMNet while maintaining convergence and accuracy, validated through extensive benchmarks against state-of-the-art R/Python implementations.

PUBLICATIONS AND PREPRINTS

C=CONFERENCE, P=PREPRINT, S=SUBMITTED

- [C.1] Yixiao Wang*, Zishan Shao*, Ting Jiang, Aditya Devarakonda. **“Enhanced Cyclic Coordinate Descent Methods for Elastic Net Penalized Linear Models,”** *NeurIPS* 2025. [Paper] [Code]
- [C.2] Ting Jiang*, Yixiao Wang*, Hancheng Ye*, Zishan Shao, Jingwei Sun, Jingyang Zhang, Zekai Chen, Jianyi Zhang, Yiran Chen, Hai Li. **“SADA: Stability-guided Adaptive Diffusion Acceleration,”** *ICML* 2025. [Paper] [Code] [Page] [Slides]
- [C.3] Yufa Zhou*, Yixiao Wang*, Surbhi Goel, Anru R. Zhang. **“Why Do Transformers Fail to Forecast Time Series In-Context?,”** *NeurIPS Workshop on What Can(t) Transformers Do?* 2025 (Oral Presentation). [Paper] [Code]
- [C.4] Yufa Zhou*, Yixiao Wang*, Xunjian Yin*, Shuyan Zhou, Anru R. Zhang. **“The Geometry of Reasoning: Flowing Logics in Representation Space,”** *ICLR* 2026. [Paper] [Code] [Dataset]
- [C.5] Zishan Shao, Yixiao Wang, Qinsi Wang, Ting Jiang, Zhixu Du, Hancheng Ye, Danyang Zhuo, Yiran Chen, Hai Li. **“FlashSVD: Memory-Efficient Inference with Streaming for Low-Rank Models,”** *AAAI* 2026. [Paper] [Code]
- [S.1] Yixiao Wang*, Hayden McTavish*, Varun Babbar*, Margo Seltzer, Cynthia Rudin. **“Near-Optimal Scalable Constant and Piecewise Regression Trees,”**
- [S.2] Yixiao Wang*, Ting Jiang*, Zishan Shao*, Hancheng Ye, Jingwei Sun, Mingyuan Ma, Jianyi Zhang, Yiran Chen, Hai Li, **“ZEUS: Zero-shot Efficient Unified Sparsity for Generative Models,”**. [Code] [Page]

* denotes co-first authorship; first-/co-first-authored works listed first.

HONORS AND AWARDS

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| • Outstanding Graduate, University of Science and Technology of China (USTC) | 2024 |
| • China Petroleum Scholarship (3 awardees, School of the Gifted Young, USTC) | 2023 |
| • Excellent Teaching Assistant (Top 3 in the University, USTC) | 2023 |
| • Silver Prize, Outstanding Student Scholarship (Top 15%, USTC) | 2022 & 2021 |
| • “Qiang Wei Feng Gong De Yu” (Diligence and Moral Conduct) Scholarship (USTC) | 2022 |
| • Excellent President of Student Club (Origami Club, USTC) | 2022 |
| • First Prize, Chinese Mathematical Competition for University Students (Top 1%) | 2022 & 2021 |
| • Bronze Prize, Outstanding Freshmen Scholarship (Top 35%, USTC) | 2020 |

TEACHING AND OUTREACH

- **Teaching Assistant: Linear Algebra B1** Mar. 2023 – Jul. 2023
University of Science and Technology of China (USTC), Hefei, China
 - Ranked **Top 3** among all teaching assistants at USTC in the university-wide TA evaluation.
 - Built the course homepage and served as the group leader, managing a course group with over 100 students.
 - Organized weekly problem-solving sessions during weekends, delivered over 40 hours of instruction served as the group leader, and engaged with more than 200 participants, showcasing strong teaching and communication skills.
 - Independently completed solution sets for over 100 end-of-chapter exercises and provided analysis and answers to all past exam papers.
- **Origami Club President** Jun. 2021 – Jun. 2022
University of Science and Technology of China
 - Designed and organized the first origami exhibition “Yue Ran Zhi Shang” (Vividly Comes to Life on Paper) in USTC, responsible for venue decoration and maintaining order for two consecutive weeks.
 - Conducted origami teaching sessions twice per month, sharing the art of origami with over 200 participants.
 - Held special origami activities, such as Origami at Christmas and spray-painting completed origami works.
- **Volunteer** Sep. 2021 – Jan. 2022
Hefei Chunyu Parent Support Center for Intellectually Disabled Children
 - Played basketball with children with autism and helped coaches maintain order in class once per week.

SELECTED COURSE PROJECTS

- **Airbnb Price Prediction in New York City** (CS 671 Theory & Algorithms of Machine Learning, Duke University) – Achieved top 5 out of 137 participants using advanced stacking strategies, combining XGBoost, LightGBM, and 20+ weak learners for robust generalization. [Report][Code]

ADDITIONAL INFORMATION

Languages: English, Mandarin, Sichuanese dialect.

Programming & Software: Python (primary), C/C++, $\text{\LaTeX}$, Linux, HTML/CSS, MATLAB, R.

Interests: Origami and other visual arts (selected works available at [portfolio link](#)).